AP Precalculus **Unit 1, Lesson 1.14**

Exit Ticket — Answer Key

Name: ____________________________ Date: ________________ Period: _________

1. A quadratic model gives the height of a projectile in feet: $f(x) = -2x^2 + 12x$, where x is time in seconds.

   (a) Evaluate $f(3)$ and $f(5)$.

   $f(3) = -2(\mathbf{3})^2 + 12(\mathbf{3}) =$ **18** ft

   $f(5) = -2(\mathbf{5})^2 + 12(\mathbf{5}) =$ **10** ft

   (b) Compute the average rate of change of f on $[3, 5]$. Include units.

   $$\text{ARC} = \frac{f(5) - f(3)}{5 - 3} = \frac{\mathbf{-8}}{\mathbf{2}} = -4$$

   Units: **feet per second (ft/s)**

2. A rational model for the force between two magnets is $P(d) = \dfrac{50}{d^2}$, where d is the distance in meters between the magnets.

   State the domain restriction and explain in one sentence why it applies.

   Domain: $d > 0$ Explanation: **Distance between physical objects must be a positive number; $d = 0$ would mean the magnets occupy the same point, which is physically impossible and causes division by zero.**

3. The average rate of change of the pen area $A(w)$ on $[6, 8]$ is -4 ft^2/ft.

   What does this tell you about the pen design on that interval? Answer in a complete sentence.

   On average, each additional foot of width between 6 and 8 feet decreases the area by 4 square feet, meaning the pen is becoming less efficient (the design is past its optimal width).

Teacher Note

Exit ticket item (1b): the negative ARC (-4 ft/s) means the projectile is losing height on average between $t = 3$ and $t = 5$ seconds, consistent with the parabola opening downward. Item (3) checks conceptual understanding of the sign and units of ARC. Collect and sort: students missing units or sign need re-teaching of ARC interpretation.